

Total No. of Questions : 10]

SEAT No. :

P1515

[5460]-106

[Total No. of Pages : 3

T.E.

(CIVIL)

Advanced Surveying

(2012 Pattern) (Semester-II) (301007)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.No. 1 or Q. No. 2, Q.No. 3 or Q. No. 4, Q.No. 5 or Q. No. 6, Q.No. 7 or Q. No. 8, Q.No. 9 or Q. No. 10.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1) a)** Define,
- i) Well conditioned triangle
 - ii) Strength of a figure
 - iii) Accuracy of triangulation
 - iv) Geodetic Surveying
 - v) Indivisibility of stations **[5]**
- b) Define triangulation, state the object of triangulation and state its applications. **[5]**

OR

- Q2) a)** Describe briefly various applications of Global Positioning System. **[5]**
- b) State any five advantages of space based positioning systems. **[5]**

- Q3) a)** Describe in brief how location survey for pier of a bridge is carried out at site. **[5]**
- b) Explain the three point problem and method of solution of three point problem using any one graphical method. **[5]**

OR

- Q4) a)** State various methods of locating the position of boat in hydrographical surveying and explain briefly. **[5]**
- i) location by two angles from Boat.
 - ii) location by one angle from shore and the other from the Boat.
- b) Describe the procedure for setting out a tunnel, explain with a sketch. **[5]**

P.T.O.

Q5) a) What do you mean by a spherical excess and how do you find out the Area of a spherical triangle? [5]

b) Define the following terms. [5]

i) Mistake

ii) True error

iii) Most probable value

iv) Conditioned equation

v) Weight of an observation

c) Following observation were recorded for an angle under identical condition. [8]

162° 20' 00" 162° 21' 20" 162° 21' 40"

162° 20' 40" 162° 19' 40" 162° 21' 20"

Calculate **a)** the most probable error of single observation,

b) the most probable error of mean,

c) the most probable value of the angles

OR

Q6) a) Explain laws of weight. [5]

b) Explain step by step procedure for figure adjustment for a geodetic quadrilateral without central station. [5]

c) Neglecting the spherical excess, adjust the angle of triangle of which observed values are [8]

Angle	Weight
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Angle A = 48° 18' 22"	3
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Angle B = 76° 32' 47.2"	1
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Angle C = 55° 08' 53.8"	3
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Q7) a) Write a stepwise procedure of determine air base distance using mirror stereoscope. [5]

b) Write short notes on: Crab and Drift [5]

c) The scale of aerial photograph is 1: 12000. The size of aerial photograph is 250 mm x 250 mm. The longitudinal overlap is 60% and side overlap is 30%. Determine the number of photographs required to cover an area of 250 sq.km. [6]

OR

Q8) a) Define the following terms.

- 1) Air base distance.
- 2) Relief displacement.
- 3) Oblique photograph.
- 4) Principal point.
- 5) Mosaic.

[5]

b) Define Ground Control Points, state their role in photogrammetry and bring out difference between pre marked and post marked Ground Control Points (GCP).

[5]

c) Find the number of photographs (size 250 x 250mm) require to cover an area of 20 km x 16 km if the longitude overlap is 60% and the side overlap is 30%. Scale of photograph is 1cm; 150m.

[6]

Q9) a) What are the components of a GIS?

[5]

b) Explain the applications of GIS in Visibility analysis and slope analysis.

[5]

c) What is GIS? State various GIS software's and explains how remote sensing and GIS are linked?

[6]

OR

Q10)a) Define remote sensing. State importance of digital image processing.

[5]

b) Discuss in brief the various data sources to build GIS for civil engineering applications such as watershed development.

[5]

c) Enlist advantages and limitations of remote sensing.

[6]

